A Two-Dimensional Framework for Assessing the Translation Quality of Academic Concepts in Social Science Texts

1. Theoretical Foundations and Framework Architecture

The translation of academic concepts in social science texts constitutes a form of terminology work, in which "concept and its relationship to other concepts are the starting point" (Felber 1984: 103). Since concepts are units of knowledge that "cannot be perceived by the senses" (Felber 1984: 167) and must therefore be carried by linguistic symbols, and since social scientific inquiry is built upon "concepts and concept formation" organized into "argumentative sequence[s]" (Ghosh 2001: 61–62), the quality of academic concept translation depends on two analytically distinct but mutually constitutive dimensions:

- **Concept Representation** — the symbolic carriage of concepts through linguistic designations (terms and proper names), governed by the principles codified in ISO 704:2022;
- **Concept Logic** — the argumentative carriage of concepts through reasoned discourse, governed by the architecture of argument (Williams 2004) and realized through figures of argument (Fahnestock 2011) and metadiscourse (Hyland 2019).

The framework, hereafter referred to as the **Concept-Based Translation Quality Assessment Framework (CB-TQA)**, integrates these two dimensions and adopts the scoring mechanism of the Multidimensional Quality Metrics (MQM) framework (Lommel, Burchardt and Uszkoreit 2014) to ensure operational comparability with established translation quality assessment practice.

CONCEPT-BASED TRANSLATION QUALITY (CB-TQA)

DIMENSION 1

Concept
Representation

Default severity:
MINOR

Felber (1984)
ISO 704:2022
Faber (2012)

DIMENSION 2

Concept
Logic

Default severity:
MAJOR

Williams (2004)
Fahnestock (2011)
Hyland (2019)

Scoring mechanism adapted from MQM
(Lommel, Burchardt & Uszkoreit 2014)

2. Dimension 1 — Concept Representation

2.1 Object of Assessment

This dimension evaluates whether the translator has selected appropriate linguistic designations to carry the source concept into the target language. Since translators of specialized language texts "must go beyond correspondences at the level of individual terms" and "establish interlinguistic references to entire knowledge structures" (Faber 2012: 9), representation quality is assessed not only at the level of the individual designation but also at the level of the concept system that the designations collectively articulate.

2.2 Sub-parameters

Code	Sub-parameter	Assessment Question	Source
CR-1	Designation type accuracy	Has the appropriate designation type (term vs. proper name) been selected?	ISO 704:2022
CR-2	Consistency	Is the same designation used consistently for the same concept across the text?	ISO 704:2022
CR-3	Derivability and compoundability	Does the designation permit derivation and compounding to express the concept system?	ISO 704:2022
CR-4	Linguistic correctness	Does the designation conform to the morphological, syntactic, and orthographic norms of the target language?	ISO 704:2022
CR-5	Preference for established target-language designations	Has the translator preferred existing target-language designations over neologisms where appropriate?	ISO 704:2022

Code	Sub-parameter	Assessment Question	Source
CR-6	Designation relations	Are mononymy, synonymy, equivalence, antonymy, and polysemy/homonymy handled appropriately?	ISO 704:2022

3. Dimension 2 — Concept Logic

3.1 Object of Assessment

This dimension evaluates whether the translator has preserved the reasoned discourse through which concepts are linked to other concepts in the source text. Following Heim and Tymowski (2006: 8–10), "altering the method of argumentation" is identified as one of the principal pitfalls of social science translation. Concept logic is assessed at two layers: the **argument schema**, which constitutes the skeleton of the argument, and the **rhetorical topology**, which constitutes its surface realization.

3.2 Layer A — Argument Schema (Williams 2004)

Code	Sub-parameter	Assessment Question
CL-A1	Claim	Has the conclusion toward which the argument converges been preserved without distortion, weakening, or reversal?
CL-A2	Grounds	Have the facts, observations, and shared knowledge supporting the claim been transferred accurately?
CL-A3	Warrant	Have the inferential statements connecting grounds to claim been preserved, including implicit warrants underlying the communicative situation?
CL-A4	Backing	Has the overarching principle, value, or standard supporting the warrant been preserved?

Note: Williams's (2004) original parameters of *qualifier/modalizer* and *rebuttal/exception* are not retained as separate sub-parameters of the argument schema, since they overlap substantively with Hyland's (2019) hedges. They are therefore subsumed under CL-B11 below, in keeping with the principle of categorial parsimony.

3.3 Layer B — Rhetorical Topology

Williams's (2004) original rhetorical topology comprises six sub-parameters (organizational relations, connectives, propositional functions, types of argument, figures, narrative strategy) whose definitions overlap and whose operationalization has proven difficult. The present framework reconstructs this layer using two more clearly defined and empirically grounded sub-layers: **figures of argument** (Fahnestock 2011), which capture the stylistic devices that structure argumentative content, and **metadiscourse** (Hyland 2019), which captures the interactive and interactional resources through which writers organize discourse and engage readers.

3.3.1 Figures of Argument (Fahnestock 2011)

Code	Figure	Realizations
CL-B1	Parallelism	(1) Comparable length of segments in syllables or words; (2) similar prosodic or stress patterns; (3) similar grammatical structure
CL-B2	Strategic repetition	(1) Repeating the openings of successive clauses; (2) Repeating the closings of successive clauses; (3) Repeating the closing of one clause in the opening of the next; (4) Repeating the opening of a clause in its ending; (5) Repeating the same openings and endings in a series of clauses; (6) Repeating a word or phrase immediately, with nothing intervening; (7)

Code	Figure	Realizations
		Repeating a word or phrase with one or a few words intervening
CL-B3	Antithesis	(1) contraries; (2) contradictories; (3) correlatives
CL-B4	Antimetabole	A – B / B – A reversal patterns
CL-B5	Definition	Subject – Linking Verb – Predicate Noun/Modifier

3.3.2 Metadiscourse (Hyland 2019)

Interactive resources (organizing the discourse for the reader):

Code	Resource	Function
CL-B6	Transition markers	Mainly conjunctions and adverbial phrases to signal addition, comparison, or consequence between argumentative steps
CL-B7	Frame markers	(1) Acting as more explicit additive relations; (2) label text stages; (3) announce discourse goals; (4)indicate topic shifts
CL-B8	Endophoric markers	Expressions which refer to other parts of the text
CL-B9	Evidentials	Invoke external sources of support
CL-B10	Code glosses	Supply additional information by rephrasing, explaining or elaborating what has been said

Interactional resources (positioning writer and reader):

Code	Resource	Function
CL-B11	Hedges (incorporating Williams's *qualifier* and *rebuttal*)	(1) Mitigate commitment; (2) admit alternative voices
CL-B12	Boosters	(1) Close down alternatives; (2) head off conflicting views; (3) express their certainty
CL-B13	Attitude markers	Convey affective stance toward propositions
CL-B14	Self-mention	Mark authorial presence (first-person pronouns and possessive adjectives, e.g. I, me, mine, exclusive we, our, ours)
CL-B15	Engagement markers	Address readers directly as discourse participants (1) addressing them as participants in an argument with reader pronouns (you, your, inclusive we) and interjections (by the way, you may notice); (2) pulling readers into the discourse at critical points through questions, directives (mainly imperatives such as see, note and consider and obligation modals such as should, must, have to, etc.)

4. Scoring Mechanism

4.1 Rationale for a Dimension-Bound Binary Severity System

The MQM framework specifies three default severity levels — minor, major, and critical (Lommel, Burchardt and Uszkoreit 2014). Although theoretically appealing, this tripartite distinction has proven operationally unstable in practice: the boundary between *major* and *critical* errors rests on the judgment of whether a text remains "fit for purpose," a criterion that is intrinsically gradient rather than categorical and that depresses inter-coder agreement. To enhance reliability while preserving the theoretical insight that the two dimensions of concept translation carry asymmetric weight in social scientific discourse, the CB-TQA framework adopts a **dimension-bound binary severity system**:

- All errors identified under the **Concept Representation** dimension are classified as **minor** (severity multiplier = 1).
- All errors identified under the **Concept Logic** dimension are classified as **major** (severity multiplier = 5).

This binding operationalizes Williams's (2004: 22–23) claim that argument constitutes "the essential elements of a text's message" in reasoned discourse: designation errors typically remain recoverable through co-text and reader inference, whereas argument errors distort the conceptual relations that constitute the knowledge claim itself. The severity multipliers (1 and 5) follow the MQM defaults derived from the LISA QA Model and are retained to ensure comparability with the wider TQA literature.

4.2 Core Formula

The overall translation quality score is calculated as follows:

$$\mathrm{TQ} = 100 - (P_{\mathrm{CR}} + P_{\mathrm{CL}}) \times 100$$

where the penalty values for each dimension are:

$$P_{\mathrm{CR}} = \frac{n_{\mathrm{CR}} \times 1}{\text{Word Count}}$$

$$P_{\mathrm{CL}} = \frac{n_{\mathrm{CL}} \times 5}{\text{Word Count}}$$

n_{CR} denotes the total number of errors recorded across the 7 CR sub-parameters;

n_{CL} denotes the total number of errors recorded across the 19 CL sub-parameters.

4.3 Error Identification Criteria — Concept Representation (CR, minor, ×1)

An error is recorded under each CR sub-parameter when the target text exhibits the following deviation from the source. All such errors carry a uniform severity multiplier of 1, reflecting the recoverability of designation-level deviations through co-text.

Code	Sub-parameter	Error criterion
CR-1	Designation type accuracy	An inappropriate designation type is selected (e.g., a term is rendered as a proper name, or a proper name as an appellative term), thereby misclassifying the conceptual category of the source designation.
CR-2	Consistency	The same source concept is rendered by varying target designations without principled motivation, or distinct source concepts are conflated under a single target designation.
CR-3	Derivability and compoundability	The selected target designation cannot accommodate the derivational or compounding patterns required to express related concepts within the source concept system.
CR-4	Linguistic correctness	The target designation violates the morphological, syntactic, or orthographic norms of the target language.
CR-5	Preference for established target-language designations	An established target-language designation that is current in the relevant scholarly community is bypassed in favour of an unwarranted neologism, calque, or borrowing.
CR-6	Transliteration and transcription	Standard transliteration or transcription conventions of the target language are misapplied or applied inconsistently.
CR-7	Designation relations	Mononymy, synonymy, equivalence, antonymy, or polysemy/homonymy in the source designation system are not appropriately reproduced in the target designation system.

4.4 Error Identification Criteria — Concept Logic (CL, major, ×5)

An error is recorded under each CL sub-parameter when the target text exhibits the following deviation from the source. All such errors carry a uniform severity multiplier of 5, reflecting the centrality of argumentative carriage to the transmission of the knowledge claim.

Layer A — Argument Schema (Williams 2004)

Code	Sub-parameter	Error criterion
CL-A1	Claim	The conclusion toward which the argument converges is distorted, weakened, intensified, reversed, or omitted in the target text.
CL-A2	Grounds	The factual evidence, observational data, or shared knowledge supporting the claim is misrepresented, attenuated, or omitted.
CL-A3	Warrant	The inferential link connecting grounds to claim — whether explicitly articulated or implicitly presupposed — is broken, altered, or rendered opaque.
CL-A4	Backing	The overarching principle, value, theory, or disciplinary standard that licenses the warrant is misrepresented or omitted.

Layer B-1 — Figures of Argument (Fahnestock 2011)

Code	Sub-parameter	Error criterion
CL-B1	Parallelism	Isocolon, parison, or prosodic patterning that establishes coordination among argumentative units is dissolved, obscuring the parity relation between conjoined claims.
CL-B2	Strategic repetition	Anaphora, epistrophe, anadiplosis, epanalepsis, symploce, epizeuxis, or diacope is dissolved, weakening the rhetorical emphasis or the chained inferential progression.
CL-B3	Antithesis	A source-text opposition based on contraries, contradictories, or correlatives is flattened, asymmetrized, or reversed, distorting the contrastive structure of the argument.
CL-B4	Antimetabole	A source-text A–B / B–A reversal pattern is dissolved or rendered as paraphrase, dissipating the conceptual chiasmus that carries the argumentative move.
CL-B5	Definition	The source-text definitional structure (Subject – Linking Verb – Predicate Noun/Modifier) that introduces or stipulates a concept is dismantled or rendered as loose paraphrase, weakening the constitutive force of the definitional act.

Layer B-2 — Metadiscourse: Interactive Resources (Hyland 2019)

Code	Sub-parameter	Error criterion
CL-B6	Transition markers	Markers signalling addition, comparison, or consequence between argumentative steps are omitted, mistranslated, or substituted such that the inferential relation between propositions is altered.
CL-B7	Frame markers	Markers that sequence, label, announce, or shift discourse stages are dissolved, disrupting the reader's orientation within the argumentative architecture.
CL-B8	Endophoric markers	References to other parts of the text are mistranslated or lost, severing the intratextual coherence on which cumulative argumentation depends.
CL-B9	Evidentials	Citations or attributions to external sources are misattributed, omitted, or rendered ambiguous, undermining the evidential basis of the claim.
CL-B10	Code glosses	Rephrasings, exemplifications, or elaborations that secure the writer's intended meaning are omitted or distorted, leaving the concept under-specified for the target reader.

Layer B-3 — Metadiscourse: Interactional Resources (Hyland 2019)

Code	Sub-parameter	Error criterion
CL-B11	Hedges (incorporating Williams's *qualifier* and *rebuttal*)	The epistemic modality of the source is altered such that tentative claims are presented as categorical, or admitted exceptions and alternative voices are suppressed.
CL-B12	Boosters	The certainty signalled by the source is attenuated, or unwarranted certainty is introduced into propositions that the source presents as provisional.
CL-B13	Attitude markers	The writer's affective stance toward a proposition (surprise, importance, agreement, doubt, etc.) is misrepresented, neutralized, or inverted.
CL-B14	Self-mention	Authorial presence (first-person reference, authorial agency) is suppressed where the source foregrounds it, or introduced where the source effaces it, altering the rhetorical positioning of the writer.
CL-B15	Engagement markers	Direct addresses to the reader (imperatives, second-person reference, rhetorical questions, inclusive *we*) are omitted or distorted, weakening the reader's positioning as participant in the unfolding argument.

4.5 Source-Text Crediting Mechanism

In line with the MQM principle that translators should be credited for improvements over the source text (Lommel, Burchardt and Uszkoreit 2014), and acknowledging Heim and Tymowski's (2006) observation that social science translators frequently confront source-text deficiencies, the framework permits bidirectional assessment:

$$\mathrm{TQ} = 100 - (P_{\text{CR-T}} - P_{\text{CR-S}}) \times 100 - (P_{\text{CL-T}} - P_{\text{CL-S}}) \times 100$$

where subscript T denotes target-text penalties and S denotes source-text penalties (computed by applying CR-1–CR-7 and CL-A1–CL-B15 to the source text in isolation, identifying lapses of consistency, derivability, designation relations, argument-schema integrity, figural coherence, and metadiscursive clarity within the source itself). Where the source text is not assessed, $P_{\text{CR-S}} = P_{\text{CL-S}} = 0.$ Under this formulation, a translation that corrects defects in the source may receive a TQ score above 100.

4.6 Worked Example

Assume a target text of 5,000 words yields the following error tally: 11 errors across the seven CR sub-parameters and 13 errors across the nineteen CL sub-parameters. The translation quality score is calculated as:

$$P_{\mathrm{CR}} = \frac{11 \times 1}{5000} = 0.0022$$

$$P_{\mathrm{CL}} = \frac{13 \times 5}{5000} = 0.013$$

$$\mathrm{TQ} = 100 - (0.0022 + 0.013) \times 100 = 98.48$$

5. Methodological Justification

The CB-TQA framework rests on three lines of justification:

Ontological justification. Concepts constitute the fundamental units of social scientific knowledge (Felber 1984; Faber 2012). Their cross-linguistic transfer therefore depends on two carriers: linguistic designations as symbolic vehicles (Felber 1984: 167; ISO 704:2022) and argumentative sequences as logical vehicles (Ghosh 2001; Williams 2004). The two dimensions of the framework correspond to these two modes of carriage.

Methodological justification. Williams (2004) provides the most developed argumentation-centred framework for translation quality assessment, but his rhetorical topology comprises six overlapping sub-parameters whose operationalization is unstable. The present framework reconstructs this layer using Fahnestock's (2011) figures of argument and Hyland's (2019) metadiscourse model — both of which are corpus-validated and widely deployed in discourse analysis — and consolidates Williams's qualifier and rebuttal under Hyland's hedges to eliminate redundancy.

Procedural justification. The differential weighting of the two dimensions (CR default minor; CL default major) operationalizes Williams's (2004: 22–23) claim that argument constitutes "the essential elements of a text's message" in reasoned discourse. The integration of escalation rules and the source-text crediting mechanism aligns the framework with the flexibility principle of MQM (Lommel, Burchardt and

Uszkoreit 2014), avoiding the rigidity that has limited the applicability of earlier one-size-fits-all metrics such as the LISA QA Model and SAE J2450.

References (for the framework section)

Faber, Pamela, ed. 2012. *A Cognitive Linguistics View of Terminology and Specialized Language*. Berlin: De Gruyter Mouton.

Fahnestock, Jeanne. 2011. *Rhetorical Style: The Uses of Language in Persuasion*. Oxford: Oxford University Press.

Felber, Helmut. 1984. *Terminology Manual*. Paris: Unesco / Infoterm.

Ghosh, Peter. 2001. "Translation as a Conceptual Act." *Max Weber Studies* 2 (1): 59–63.

Heim, Michael Henry, and Andrzej W. Tymowski. 2006. *Guidelines for the Translation of Social Science Texts*. New York: American Council of Learned Societies.

Hyland, Ken. 2019. *Metadiscourse: Exploring Interaction in Writing*. 2nd ed. London: Bloomsbury.

ISO. 2022. *ISO 704:2022 — Terminology Work — Principles and Methods*. Geneva: ISO.

Lommel, Arle, Aljoscha Burchardt, and Hans Uszkoreit. 2014. "Multidimensional Quality Metrics: A Flexible System for Assessing Translation Quality." In *Proceedings of Translating and the Computer 35*. London: Aslib.

Williams, Malcolm. 2004. *Translation Quality Assessment: An Argumentation-Centred Approach*. Ottawa: University of Ottawa Press.